

Project Helix: Building Dual Engines for Self-Sustaining Growth

- Transform into a commercial-stage biopharma leader within 36 months through a dual-track strategy: malaria vaccine + world-class CDMO.

- Next-generation malaria vaccines have created a historic opportunity to serve a vast, unmet global health need.
- The biologics CDMO market shows strong secular growth, driven by a structural deficit of high-quality, globally compliant manufacturing capacity.

- Track 1 (Vaccine): In-license a world-class malaria vaccine (2027 launch) to build a high-growth, high-margin portfolio and position as a global health leader.
- Track 2 (CDMO): Invest in a globally compliant facility to capture biologics growth, generate stable revenue, and de-risk the enterprise.

Our Vision

The Opportunity

Our Dual-Track Strategy

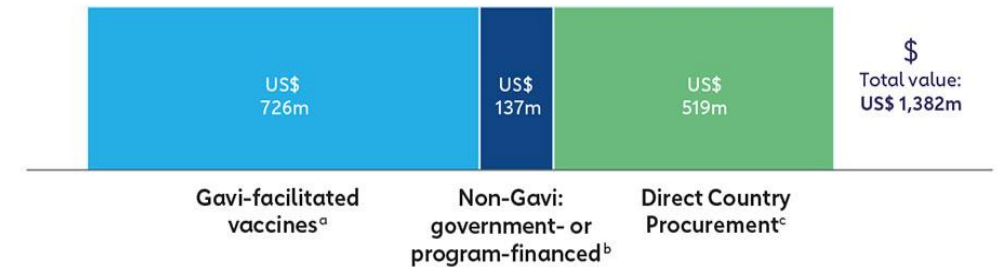
Problem: Africa's Vaccine Security Gap vs. VaccineCo's Constraints

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- Africa consumes 25% of global vaccines but **imports 99%** of them;
While market size from **\$1.3B (2025)** to **\$4.7B (2030)**
- **Drug substance (DS) = bottleneck**: only **61M local doses** by 2030 vs. **1.4B needed**
- Procurement is **price-driven** (Gavi/UNICEF), limiting margins and visibility
- Logistics & regulatory barriers (cold-chain gaps, fragmented approvals) slow expansion
- **VaccineCo today**: strong **fill-&-finish** + tech transfers, but **weak DS capacity**

Conclusion: VaccineCo can't scale without solving **demand certainty & financing for DS facilities**.

Figure 8 Market share of Gavi-supported vaccines in Africa by value, 2020 (million US dollars)



^a UNICEF Supply Division, Procurement on behalf of Gavi-facilitated on shipment end date in 2020. Gavi-facilitated doses include doses financed by governments for fulfilment of co-finance requirements.

^b UNICEF Supply Division, Procurement on behalf of other programs and governments, based on shipment end date in 2020

^c This figure is significantly driven by vaccine procurement by North African countries and South Africa.

Source: UNICEF, Linksbridge, WHO Market Information for Access to Vaccines (MI4A) and CHAI 2022- UNICEF procurement records based on reported shipment end date, 10 Oct 2022

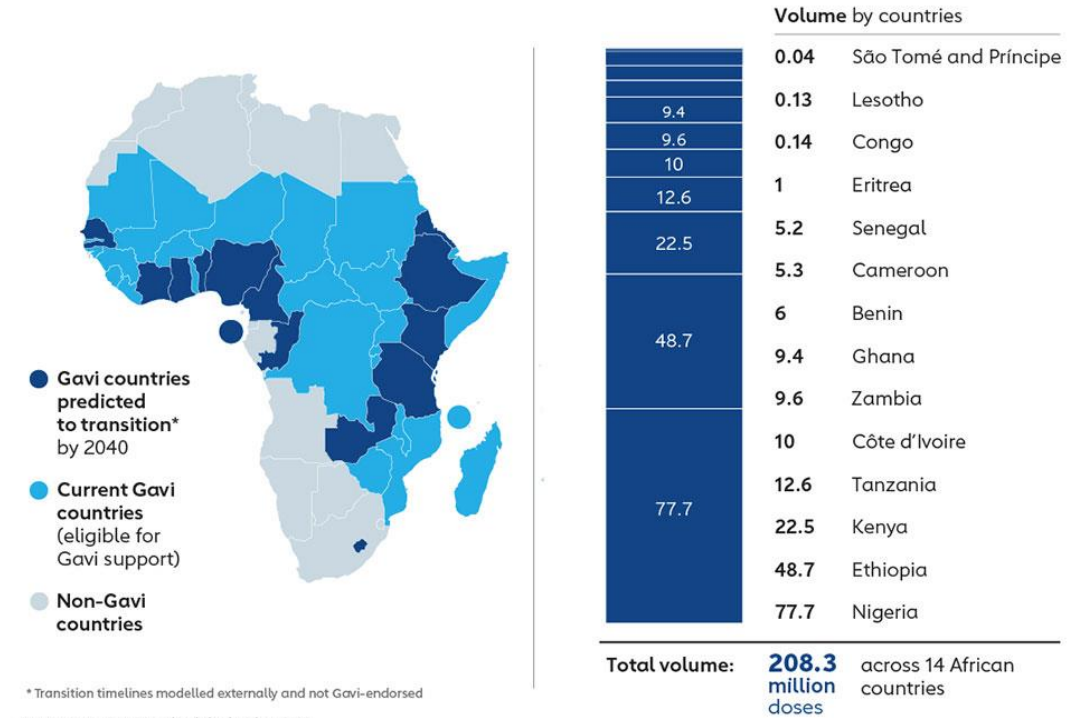
"Africa's vaccine security gap is not only supply-side (imports vs. local production), but also financing-side (donor-driven vs. sustainable local procurement)"

Resources: [resource link](#)

Strategic Diagnosis - Why Scale-Up Isn't Financeable Today

- **Demand uncertainty** → Gavi/UNICEF = price-driven, short-term tenders → violate revenue
- **Capability mismatch** → Africa's base = F&F heavy, DS scarce → poor control of COGS, IP, yields
- **System friction** → cold-chain gaps + fragmented regulation → long PQ/approval cycles
- **Implication for VaccineCo** = under-utilized assets + uncertain cash flows → **DS capex blocked**

Figure 10 Vaccine doses supported by Gavi in 2020, in countries predicted to transition by 2040 (million doses)



"Dependence on Gavi today is high, but transitions by 2040 will force African producers to build autonomy → VaccineCo needs to act now."

Opportunity Identification - Two Unlocks (Dual-Track Strategy) for VaccineCo

A. Anchor Product (Malaria Vaccine)

- Secure exclusive malaria tech-transfer + offtake with Gavi/UNICEF
- Collateralize DS financing through **AVMA (\$1.2B)** + **IFC blended loans**

B. Operations Upgrade (WHO Prequalification)

- Run PQ track in parallel → unlock bigger buyer pool, better pricing power
- Align with AU policies (*Buy African*, pooled procurement/AMCs)

Why now: DS gap is acute + policy tailwinds rising + new tech-transfer windows (mRNA hubs)

Figure 14 Potentially beneficial markets for new entrants against market shaping priorities

A Antigens with clear market health need for additional (e.g., "regional") manufacturers	Cholera	Measles-Rubella	Yellow Fever
	Ebola	Malaria	
B Antigens needing enhanced product profiles	Rotavirus	Pneumococcal	HPV
	Meningitis	COVID-19	
C Antigens where a new supplier would not improve / may deteriorate overall market health	Pentavalent	Polio	Typhoid Conjugate
	Japanese Encephalitis	Measles	
D Diseases without a licensed vaccine	HIV	RSV	Tuberculosis (Novel)
	Chikungunya	Lassa	Rift Valley Fever
	Nipah	MERS	Others...

"VaccineCo can become Africa's anchor vaccine producer by combining a malaria tech-transfer with WHO prequalification—balancing short-term bankability with long-term autonom."

Resources: [resource link](#)

Dual Track Strategy: Building Toward Leadership

Where we are today:

- Strength in fill & finish but no upstream drug-substance capacity
- Dependent on donor/UNICEF procurement; limited bargaining power
- Regional footprint modest; credibility gap without WHO PQ

Where we are heading

- Malaria vaccine tech transfer: (transformational anchor)
- WHO PQ + fill & finish scale-up: (cash flow +credibility)
- PQ Pivot: Once WHO PQ is achieved, pivot decision based on malaria partnership outcome

Why Dual-Track?

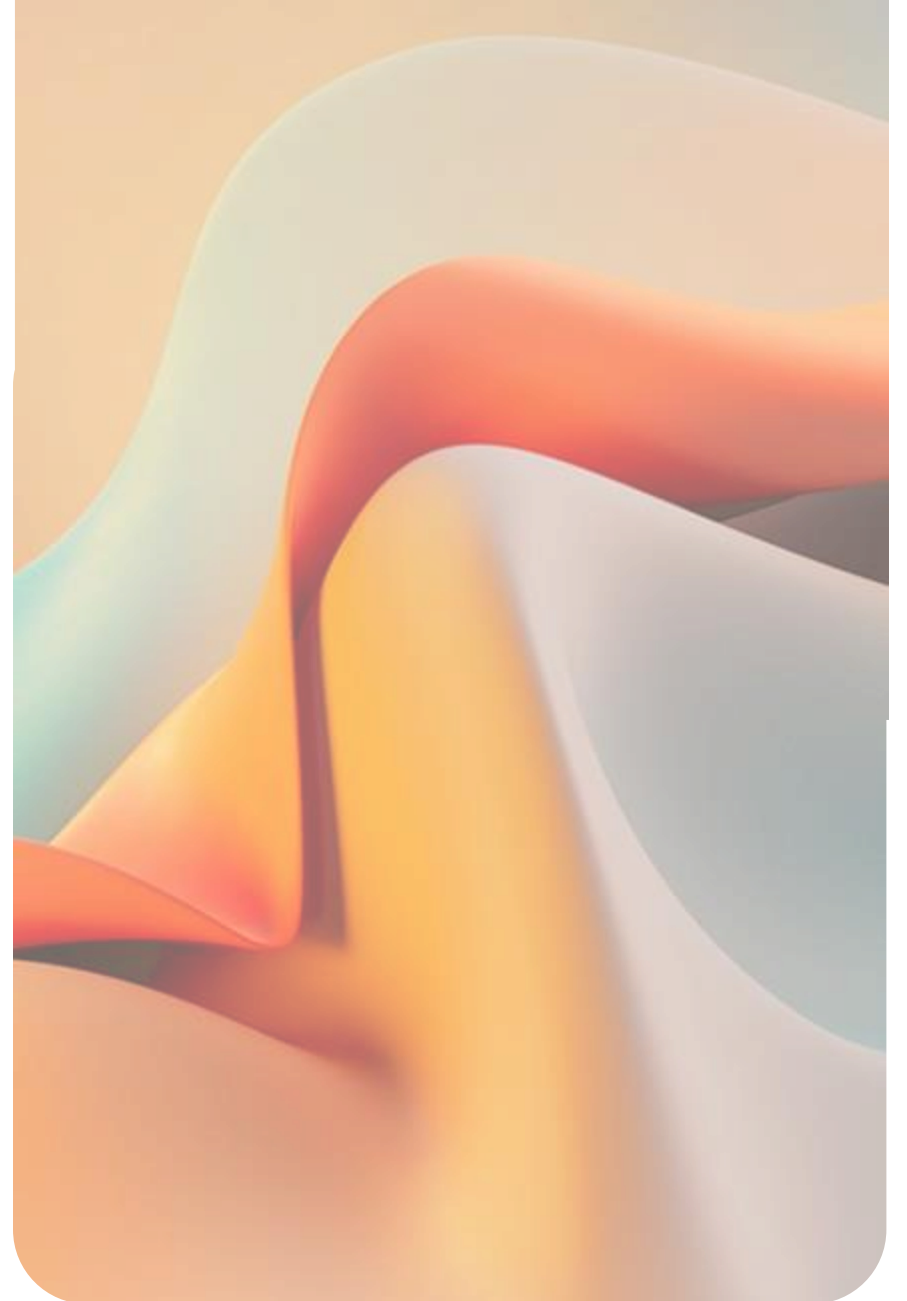
Malaria Track

- Huge, predictable demand (up to 100M doses by 2030)
- Gavi funding + rollout already underway
- Africa's most urgent health and economic need

CDMO / Fill & Finish Track

- Stable revenues, faster WHO PQ credibility
- Leverages existing assets, builds resilience
- Flexible platform for global partners

Together: Breakthrough leadership + financial stability



Initiative 1 : Malaria Vaccine Technology Transfer

Phase 1: Value Proposition (0-6 months)

- Document previous successful partnerships (J&J/Pfizer) and IFC endorsement
- Establish Diplomatic Channels with South African government endorsement
- Target partners –
 - Oxford Serum Institute of India
 - GlaxoSmithKline

Phase 2: Technical Negotiation (6-12 months)

- Partners assess VaccineCo. Facilities – leverage COVID infrastructure
- Negotiate Commercial Terms : Royalty Rates, production, regional scope
- Obtain SADC Market access and strategize for regulatory fast-tracking

Phase 3: Finalize Agreements (12-8 months)

- Legal Documentation - binding technology transfer contract
- Financing Coordination - IFC and Afreximbank project financing (\$50 M+ facility investment)
- Implementation Design - for updating facilities and obtaining equipment

Milestones:

1. Signed Letter of Intent

2. Memo of Understanding

3. Signed Binding Agreement

Resource Requirements:

Leadership:

- Negotiation with Financial, Technology transfer, and market access partners

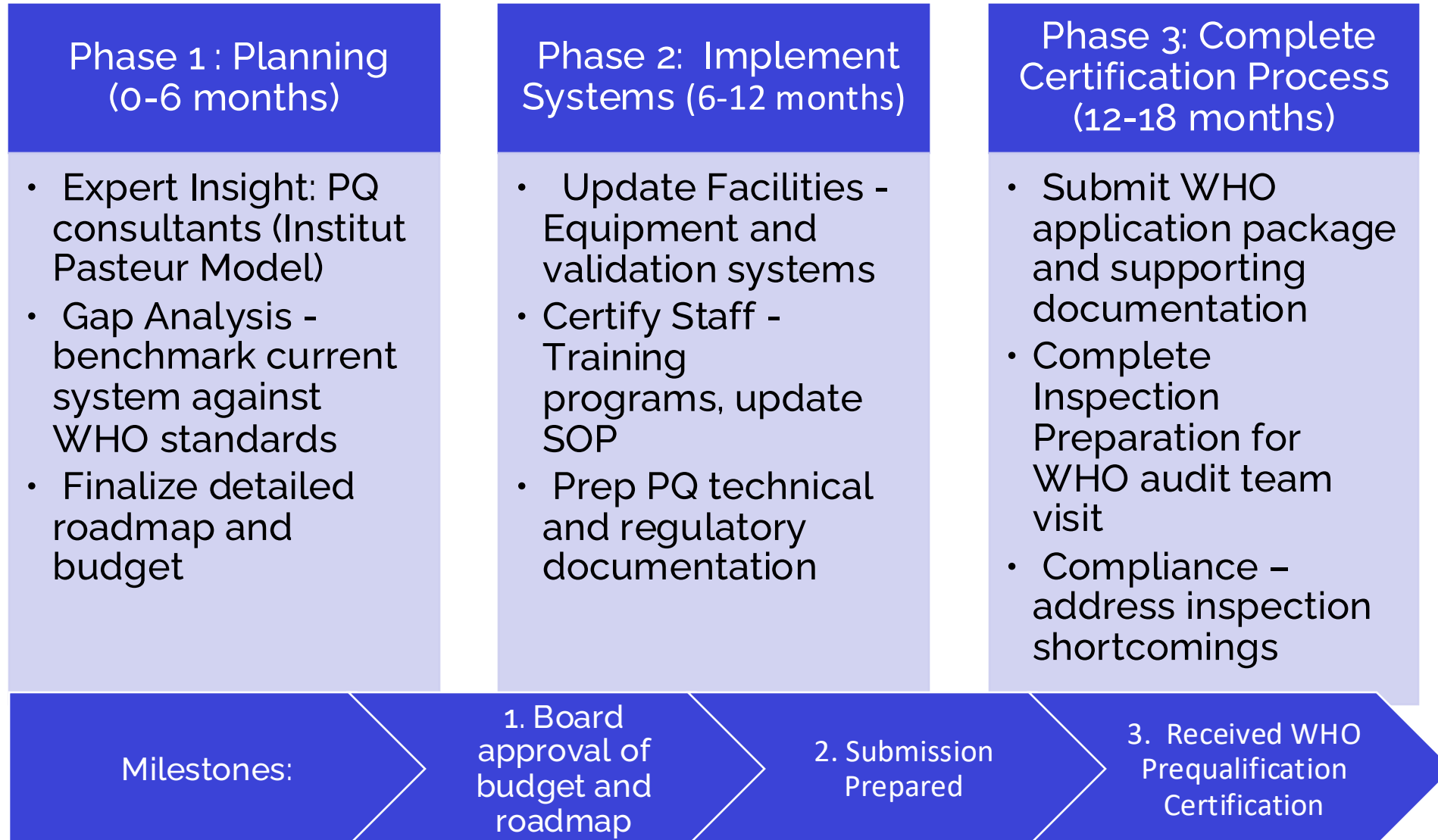
Technical Teams:

- Need of technical expertise for vaccine technology transfer

Investment:

- Up to \$5M upfront costs (defined by final agreement structure)

Initiative 2 : WHO Pre-Qualification (PQ)



Resource Requirements:

Consultants:

- ~\$200-300K for international WHO PQ consulting

Facility upgrades:

- Equipment and facility enhancements (~\$2-3 M capital investment)

Internal Teams:

- Quality and regulatory staff

Cross-Initiative Coordination

Optimization of Resources

Shared Technical Staff - Intersection between malaria technology transfer and WHO PQ process

Facility Upgrades - WHO upgrades will strengthen Malaria partnership value partnership

Timeline Intersection - both initiatives fit within an 18-month completion

Decision Protocol

Month 6: Determine Resource allocation based off Malaria Vaccine Letter of Investment success

Month 12: Final push based off both initiatives progress

Month 18: Realign VaccineCo. Strategy based off both initiative outcomes.

Investment Thesis & Capital Needs

WHO Prequalification Core Investment

Cost Range (USD)	\$8M–12.5M
Facility upgrades	\$5M–8M
Quality system setup	\$2M–3M
Certification & audit	\$1M–1.5M

>70%

can be reused for malaria vaccine projects

Malaria Vaccine Tech Transfer (Phased Light-Asset Strategy*)

Phase 1: Tech upgrade (fill-finish breakthrough)

Cost Range (USD)	\$6M–9.5M
Licensing fees	\$3M–5M
Recombinant protein equipment	\$2M–3M
Process validation	\$1M–1.5M

Oxford Univ. Matrix-M™ adjuvant licensing

Bioreactor / mixing system (critical light-asset)

3-batch consistency studies (incl. accelerated stability)

Phase 2: Drug substance capacity

Independent DS production line (biosafety required); final fill-finish line 100% reusable for PQ

\$10M–15M
(to be initiated as demand arises)

Diversified Financing Strategy



Leverage public-private status to secure low-cost, long-term debt from Development Finance Institutions (DFIs) for infrastructure upgrades, de-risking the foundation.



Target non-dilutive funding from global health partners (e.g., Gates Foundation, CEPI) to specifically finance technology transfer and WHO prequalification, accelerating key initiatives.



Utilize innovative finance like Advance Market Commitments (AMCs) and Gavi's AVMA incentives to transform market uncertainty into predictable revenue streams.



With a de-risked model, attract specialized "smart money" (Venture Capital/Private Equity) to fuel commercial scale-up, market expansion, and long-term profitability.

Financial Inflection & Growth Outlook

Strategic Investment Phase

Commercial Growth Phase

2026

2027

2028

CDMO: Facilities under construction and validation; no commercial production allowed before GMP approval.

Vaccine: Late-stage tech transfer and validation; application submitted, not yet approved.

CDMO: \$1,500,000 per batch × 10 batches = **\$15,000,000**

Vaccine: Successful launch, early stage of channel building.
2.5 million doses × \$4 per dose = **\$10,000,000**

\$96M

\$38M

Anchored client production enters a mature and stable phase, with annual delivery rising to 15 batches.

\$1,500,000 per batch × 25.3 (rounded adjustment) ≈ **\$38,000,000**

Vaccine revenue: Production supply stabilized, with approvals from more countries, achieving 20% market penetration.
Total market demand (120 million doses) × 20% (penetration) = 24 million doses
24 million doses × \$4 per dose = **\$96,000,000**